

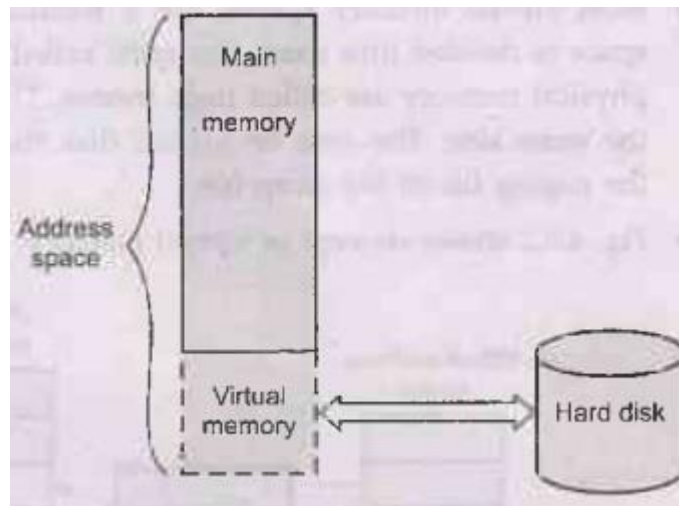
Physical Memory vs Virtual Mmemory

- Physical memory is the actual RAM chips installed in a computer, which are fast and used for actively running programs.
- Virtual memory is a technique that uses disk space (like an SSD) to extend the physical RAM, allowing the system to run more programs than it could with physical RAM alone, but it is significantly slower because it relies on slower storage.

Aspect	Physical Memory (RAM)	Virtual Memory
Definition	Actual hardware memory (RAM chips).	Illusion of a large, continuous memory space provided to each process.
Size	Limited by installed RAM.	Much larger — uses RAM + disk (swap space).
Managed by	Hardware (RAM controller).	Operating System + Memory Management Unit (MMU).
Access Speed	Very fast.	Slower when swapping with disk.
Example	8 GB physical RAM.	A process “sees” 4 GB virtual memory (even if RAM < 4 GB).

Virtual Memory

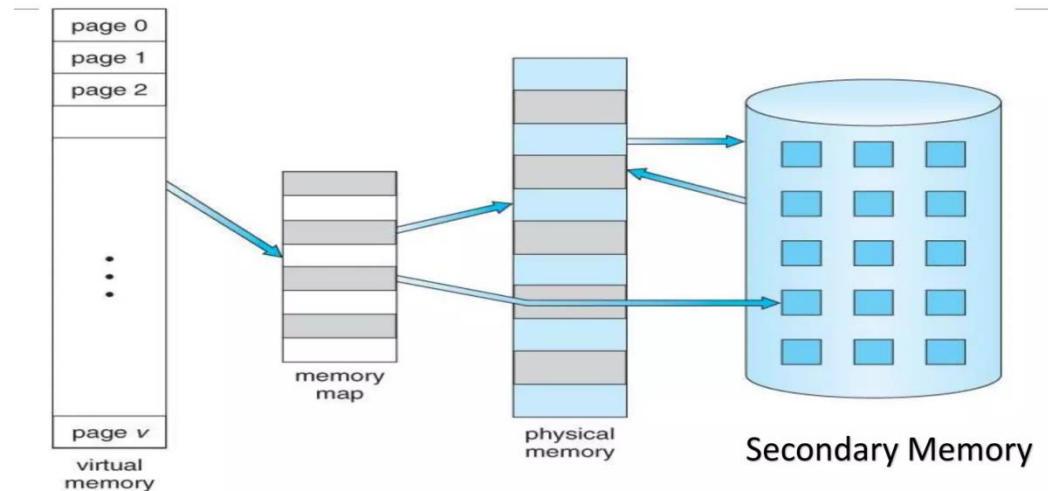
- Virtual memory is a memory management technique used by operating systems to give the appearance of a large, continuous block of memory to applications, even if the physical memory (RAM) is limited and not necessarily allocated in contiguous manner.
- The main idea is to divide the process in pages, use disk space to move out the pages if space in main memory is required and bring back the pages when needed.



Logical view of Virtual Memory

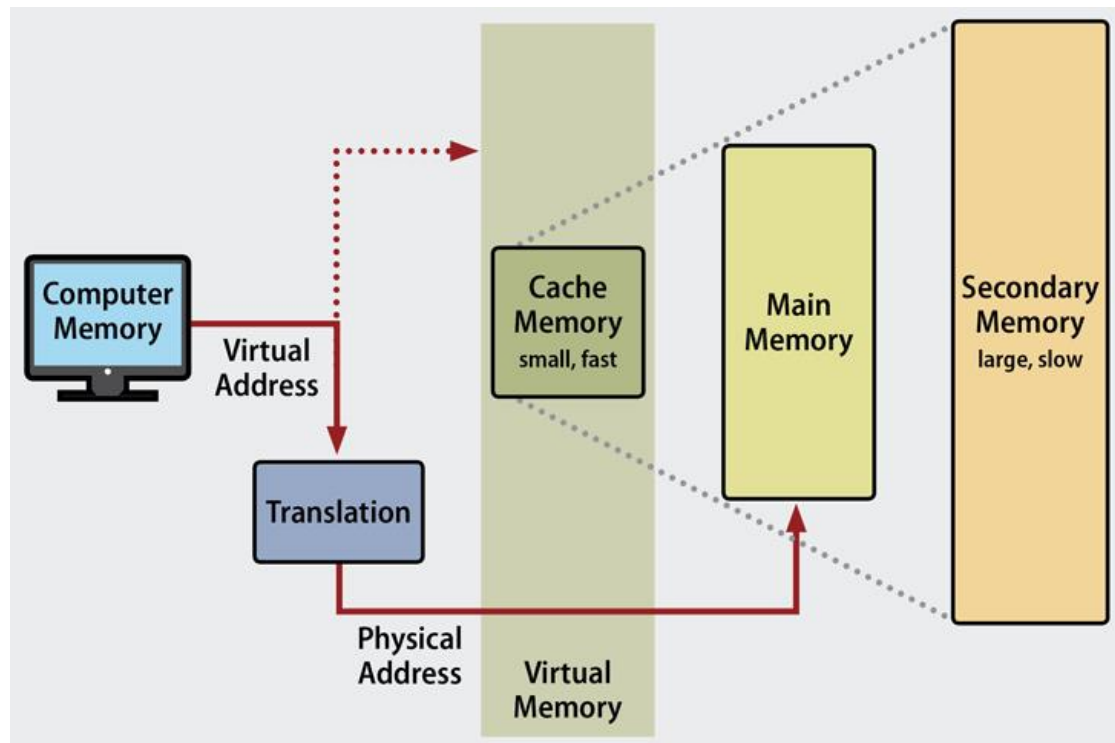
Objectives of Virtual Memory

- A program doesn't need to be fully loaded in memory to run. Only the needed parts are loaded.
- Programs can be bigger than the physical memory available in the system.
- Virtual memory creates the illusion of a large memory, even if the actual memory (RAM) is small.
- It uses both RAM and disk storage to manage memory, loading only parts of programs into RAM as needed.
- This allows the system to run more programs at once and manage memory more efficiently.



How Virtual Memory Works

- Virtual memory uses both hardware and software to manage memory.
- When a program runs, it uses virtual addresses (not real memory locations).
- The computer system converts these virtual addresses into physical addresses (actual locations in RAM) while the program runs.



Types of Virtual Memory

In a computer, virtual memory is managed by the Memory Management Unit (MMU), which is often built into the CPU. The CPU generates virtual addresses that the MMU translates into physical addresses.

There are two main types of virtual memory:

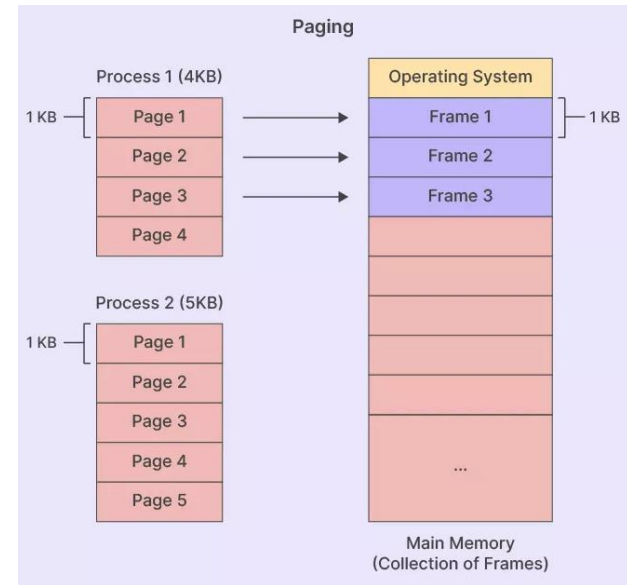
- Paging
- Segmentation

Paging: a process address space is broken into fixed-sized blocks called pages.

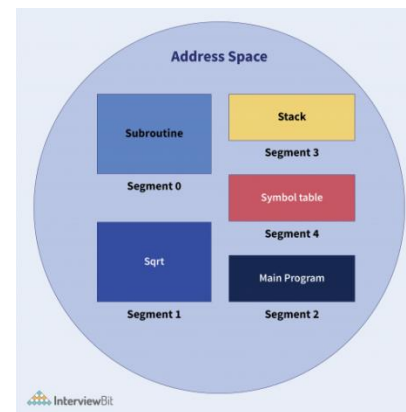
Segmentation: a process address space is broken into varying-sized blocks called sections.

Operating System divides the memory into pages.

- Paging divides the program into fixed-size pages. whereas, Segmentation divides the program into variable size segments. Paging is faster than segmentation.
- During segmentation, a logical address is divided into section number and section offset.



Segmentation



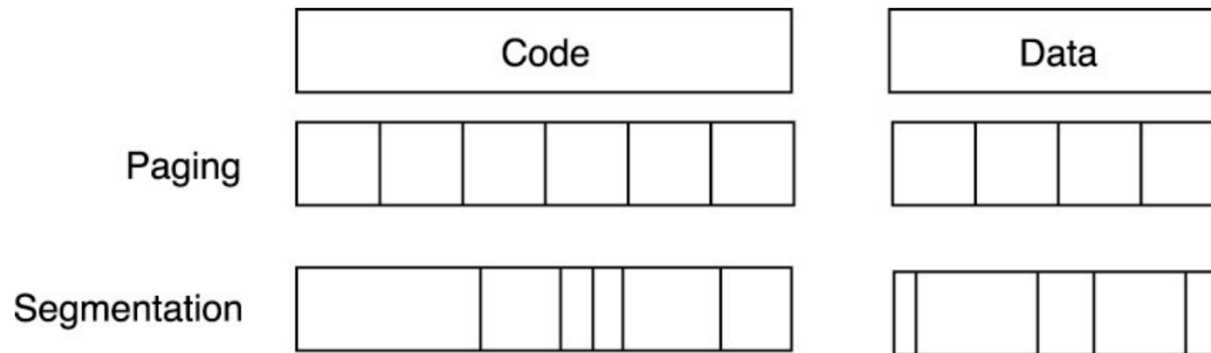
Paging vs Segmentation

S.N.	Paging	Segmentation
1	Non-Contiguous memory allocation	Non-contiguous memory allocation
2	Paging divides the program into fixed-size pages.	Segmentation divides the program into variable size segments.
3	Paging is faster than segmentation	Segmentation is slower than paging
4	Paging is closer to Operating System	Segmentation is closer to User
5	It suffers from internal fragmentation	It suffers from external fragmentation
6	The logical address is divided into a page number and page offset	The logical address is divided into segment number and segment offset
7	The page table is used to maintain the page information.	Segment Table maintains the segment information
8	Page table entry has the frame number and some flag bits to represent details about pages.	Segment table entry has the base address of the segment and some protection bits for the segments.

Internal fragmentation is wasted space within a fixed-size memory block, while external fragmentation is the unused space between variable-sized memory blocks that is too small to be allocated

Paging vs Segmentation

- Paged Virtual Memory
 - memory divided into fixed sized pages
 - each page has a base physical address
- Segmented Virtual Memory
 - memory is divided into variable length segments
 - each segment has a base physical address + length



Advantages of Paging in OS

- The memory management algorithm is easy to use.
- Swapping between equal-sized pages and page frames becomes easier.

Disadvantages of Paging in OS

- Internal fragmentation.
- Additional memory consumption by Page tables.
- Memory reference overhead due to multi-level paging.

Advantages of segmentation in OS

- Provides protection within segments.
- Segments referencing multiple processes can help achieve sharing.
- No internal fragmentation
- As compared to paging, segment tables use lesser memory.

Disadvantages of Segmentation in OS

- The separation of free memory space into small pieces can cause external fragmentation.
- It is costly.

Virtual memory organization

The Physical Address Space is conceptually divided into a number of fixed-size blocks, called **frames**.

The Logical address Space is also split into fixed-size blocks, called **pages**.

Page Size = Frame Size

Let us consider an example:

Physical Address = 12 bits, then Physical Address Space = 4 K words

Logical Address = 13 bits, then Logical Address Space = 8 K words

Page size = frame size = 1 K words (assumption)

Virtual memory organization

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